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LEC16 - Trees

Tree Terminology

- A collection of values (nodes) with directed edges between them
- A tree is either empty, or non-empty
- Every non-empty tree has a root connected to 0 or more subtrees
- The parent of a tree value (node) is the value (node) immediately above and connected to it
- The children of a tree value are the values directly connected underneath it
- The descendents of a tree value are its children, children of children… etc.
    - Can be defined recursively as its children + descendants of its children
- The ancestors of a value are its parent, parent of parent… etc.
    - Can be defined recursively as its parent + ancestors of its parent
- A path is a sequence of values $n_1, n_2, \ldots, n_k$ where there is an edge between each pair of $n_i - n_{i+1}, i < k$
- The length of a path is the number of edges in it
- There is a unique path from the root of a tree to each node in the tree
- There are no cycles (paths that form loops)

Leaf: A value with no children

Internal Node: A value with one or more children

Height of a tree: The longest path from the root to one of the leaves
- Count the number of values on the path (not the edges)
- Can be expressed as 1 + the maximum path length in a tree
- A values height is 1 + the maximum path length of the tree rooted at that node

Depth of a value: The length of the path from the root to that value
- The root itself has depth 0

Arity / Branching Factor: The maximum number of children for any node

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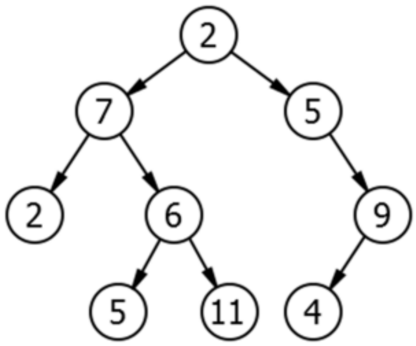
Example



Consider the following tree, find…

1. The root of this tree
    - 2
2. The parent of node 6
    - 7
3. The children of node 7
    - 2, 6
4. The leaves
    - 5, 11, 4
5. The internal nodes
    - 7, 5, 6, 9
6. A possible subtree
    - A tree with root 7 and all its descendants
7. An example path
    - 2 → 7 → 6 → 11
8. The height of the tree
    - 4 (2 → 7 → 6 → 11)
9. Depth of the values 9, 5 ,7
    - 9 → 2
    - 5 (Parent 2) → 1
    - 5 (Parent 6) → 3
    - 7 → 1
10. The arity / branching factor
    - 2 (From node 6)

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Tree Attributes and Recursive Template

```
class Tree:
      _root: Optional[Any]
      _subtrees: list[Tree]


def recursive_method(self) -> ...:
      if self.is_empty():
            ...
      if self._subtrees == []: # Not needed, but can make implementation easier
            ...
      else:
            for subtree in self._subtrees:
                  ... subtree.recursive_method() ...
      ...
```